

Competition and Inflation under HSA Demand

Nine-Cell State-Space Implementation Report

FINAL PRODUCTION RUN: FAST STATE DEMOTED

Abstract

This report is generated by the executable implementation of the design in `design.tex`. It estimates the pre-declared 3×3 price-activity matrix using a measurement-only business-demography cut, paired quarterly annualized and exactly aggregated year-over-year likelihoods, and a secondary full-joint information-flow sensitivity. The displayed run is explicitly labeled above. This final production report combines the converged core run with pre-declared measurement, persistent-error, low-frequency, and slow-only follow-up analyses.

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1 Status and interpretation boundary

The primary coefficients are conditional varying-coefficient associations. The code does not relabel observed firm counts as theoretical effective varieties, does not equate aggregate inverse markup with structural marginal cost, and does not treat full-joint feedback as an endogeneity correction. Missing benchmark-admissibility conditions remain visible as failed or conditional gates.

2 Data and executable design

The common sample contains 124 quarterly endpoints from 1982Q1 through 2012Q4. Inflation is $400\Delta \log P_t$ for PPI, headline CPI, and core CPI. The three activity measures are the log inverse-markup proxy, the frozen BN output gap, and the negative unemployment gap. Annual firm counts enter at Q4 only. Quarterly QCEW establishments provide the business-demography indicator. All nine cells share one complete-case sample; cell-specific dropping is forbidden by the loader.

The primary state posterior is the modular cut

$$p_{cut}(S, \beta_c) = p(S \mid C)p(\beta_c \mid \pi_c, X_c, S).$$

The full-joint E2 result is secondary. Its slow and fast state paths are sampled with the two conditional FFBS blocks implied by the bilinear term. For YoY, the entire quarterly conditional mean is premultiplied by A_4 and the likelihood uses $G_4 = A_4 A_4'$.

3 Nine-cell primary coefficients

Table 1: Nine-cell QoQ modular-cut E2 coefficients

Cell	Inflation	Activity	ψ	κ_0	κ_1	θ_0	γ
1	PPI	Inverse markup	0.019 [-1.305, 1.345]	-6.411 [-33.502, 21.314]	1.314 [-5.337, 8.263]	0.019 [-0.792, 0.816]	-0.016 [-0.344, 0.319]
2	PPI	BN output gap	-0.520 [-1.447, 0.535]	0.999 [-0.254, 2.237]	-0.246 [-0.702, 0.245]	0.108 [-0.707, 0.924]	-0.018 [-0.340, 0.311]
3	PPI	Negative unemployment gap	-0.276 [-1.261, 0.736]	0.100 [-0.816, 1.006]	0.072 [-0.333, 0.491]	0.079 [-0.765, 0.894]	0.003 [-0.334, 0.349]
4	CPI	Inverse markup	0.081 [-1.144, 1.298]	1.188 [-13.372, 16.173]	0.596 [-3.819, 4.850]	0.026 [-0.620, 0.694]	-0.075 [-0.323, 0.185]
5	CPI	BN output gap	-0.220 [-0.810, 0.338]	0.439 [-0.046, 0.920]	-0.117 [-0.373, 0.072]	0.153 [-0.521, 0.858]	-0.098 [-0.348, 0.184]
6	CPI	Negative unemployment gap	-0.087 [-0.617, 0.436]	0.199 [-0.184, 0.600]	0.088 [-0.106, 0.308]	0.112 [-0.529, 0.744]	-0.014 [-0.277, 0.253]
7	Core CPI	Inverse markup	0.330 [-0.701, 1.312]	1.479 [-6.751, 10.068]	0.402 [-3.007, 3.567]	0.131 [-0.353, 0.598]	-0.058 [-0.251, 0.157]
8	Core CPI	BN output gap	0.201 [-0.158, 0.605]	0.009 [-0.185, 0.196]	0.017 [-0.097, 0.114]	0.099 [-0.364, 0.562]	-0.049 [-0.227, 0.164]
9	Core CPI	Negative unemployment gap	0.273 [-0.185, 0.779]	0.061 [-0.204, 0.315]	0.071 [-0.078, 0.202]	0.122 [-0.331, 0.559]	0.008 [-0.184, 0.214]

Notes: posterior mean and central 95% credible interval. The production run uses retained post-warmup draws; convergence and identification gates are reported separately.

4 Identification before interpretation

Gate / evidence	Production status	Interpretation
Baseline quarterly information $R_q \leq 0.80$	1.086	Fail; the stationary state is not substantively interpreted.
Alternative measurement specifications	0/5 pass	The fast state is retained only if every frozen specification passes.
Fast-state decision	Demote	θ_0 and γ are reported but demoted; slow-only results are the fallback.
Baseline coefficient convergence	72/72 pass	Production $\hat{R}$ /ESS gates evaluated on retained draws.
Inflation-error robustness	107/107 mechanism pass	AR(1) and low-frequency specifications are executed across E0–E2; 5 non-mechanism intercept draws are flagged.
Activity endogeneity	Not identified	Coefficients remain conditional associations.
Chib Bayes factors	Not run	No Bayes-factor claim is made from the modular cut.
Real-time LFO score	Inadmissible	Required vintages and release rules remain unavailable.

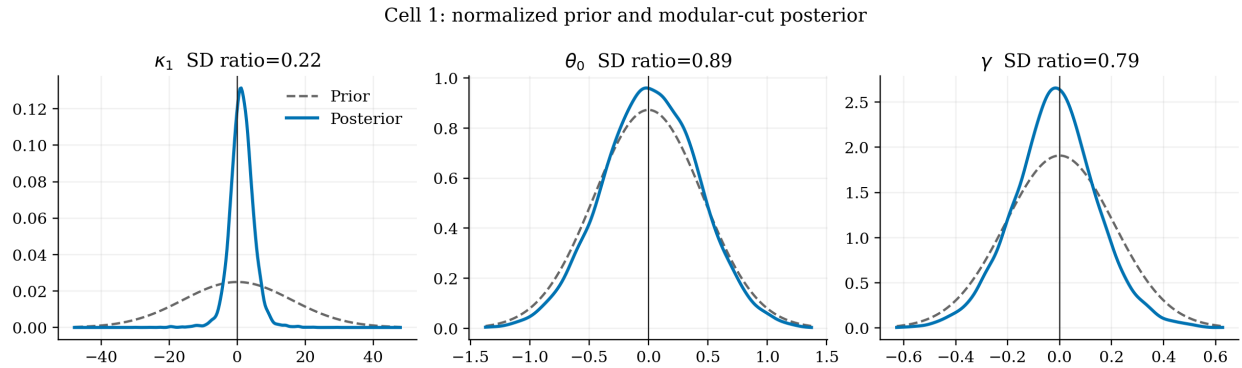


Figure 1: Cell 1 prior and primary modular-cut posterior. Production posterior summaries use retained post-warmup draws.

5 Competition-state measurement

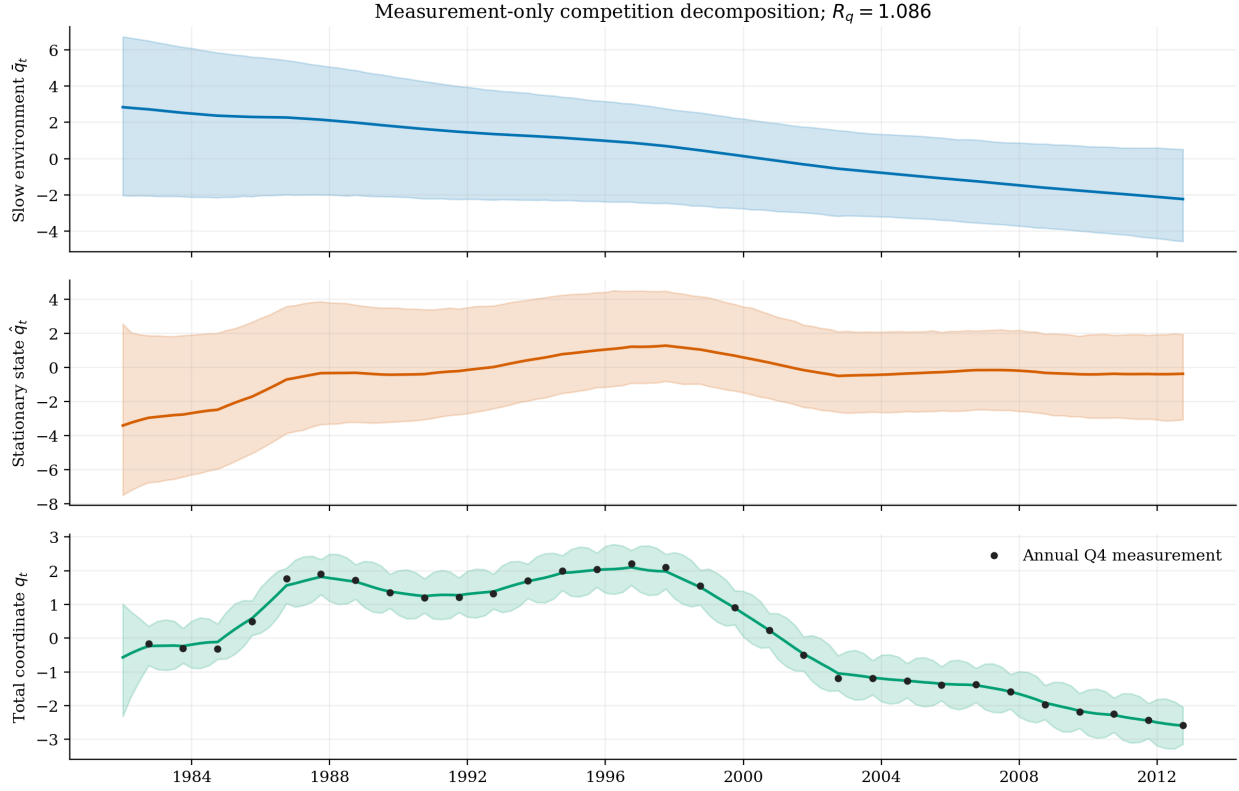


Figure 2: Measurement-only slow, stationary, and total business-demography coordinate.

5.1 Measurement-gate diagnosis and alternative state laws

Specification	R_q	Reduction	ϕ_q	b_E	Info gate	MCMC
baseline	1.086	-8.6%	0.922	0.103	Fail	Pass
quarterly no drift	1.103	-10.3%	0.963	0.291	Fail	Flag
all no drift	1.044	-4.4%	0.749	0.051	Fail	Flag
persistent cycle	1.029	-2.9%	0.949	0.108	Fail	Pass
idiosyncratic qcew cycle	1.005	-0.5%	0.918	0.133	Fail	Flag

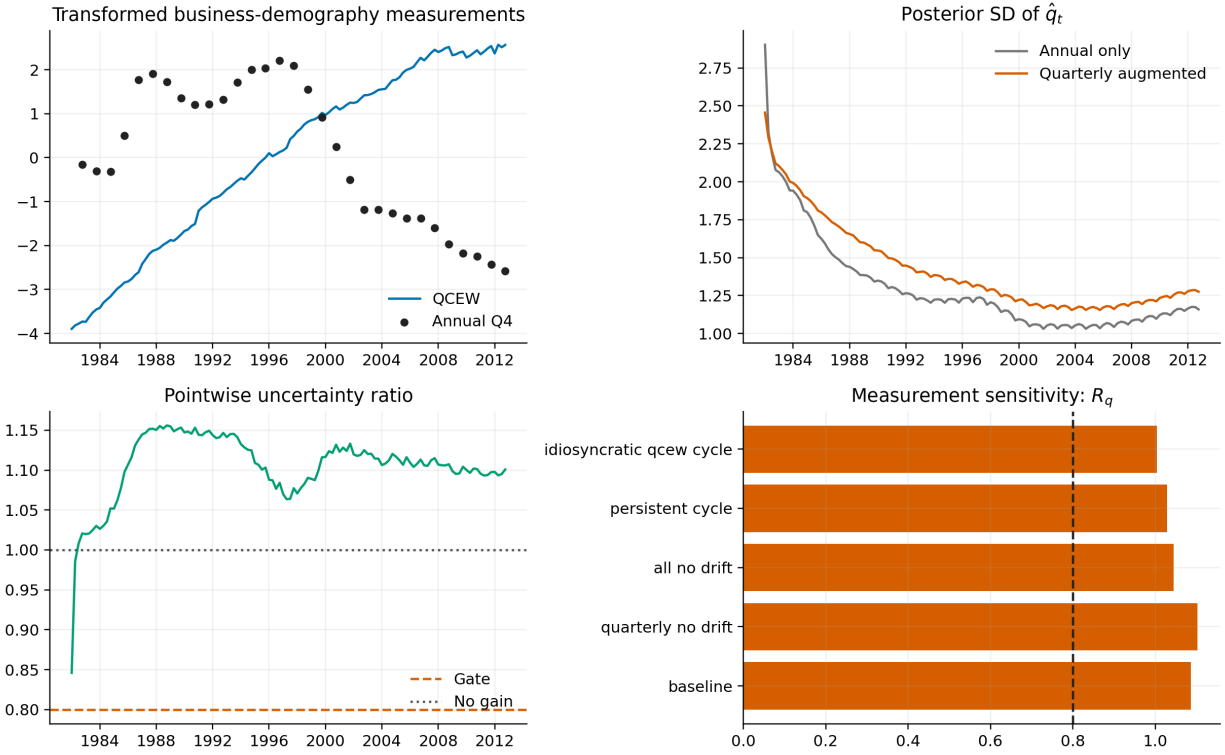


Figure 3: Observed-series alignment, annual-only versus augmented uncertainty, and the frozen alternative measurement specifications. The fast state is retained for substantive interpretation only when every displayed specification satisfies the pre-declared information gate.

6 Time-varying slope

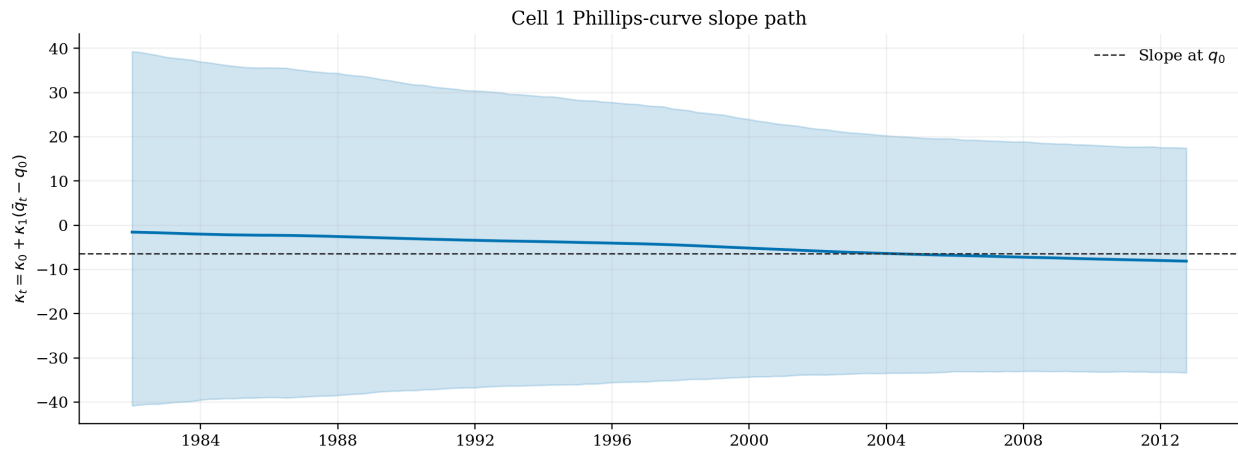


Figure 4: Cell 1 Phillips-curve slope path under the primary cut E2 model.

7 Cut versus full joint

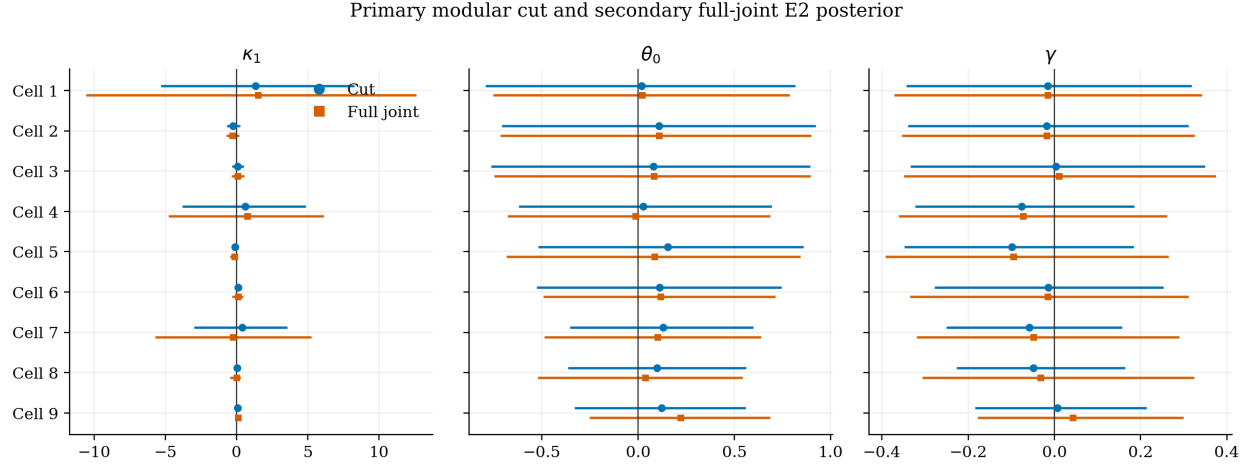


Figure 5: Primary cut and secondary full-joint posterior summaries.

Cell	Maximum shift (cut SD)	Conflicts	Status
1	0.06	0	Pass
2	0.13	0	Pass
3	0.06	0	Pass
4	0.12	1	Flag
5	0.35	0	Pass
6	0.16	0	Pass
7	0.40	1	Flag
8	0.58	1	Flag
9	0.44	0	Pass

8 Paired QoQ and exact YoY transformation

Table 2: Endpoint-matched QoQ and exactly aggregated YoY comparison

Cell	Inflation	Activity	κ_1 QoQ / YoY	θ_0 QoQ / YoY	γ QoQ / YoY
1	PPI	Inverse markup	1.231 [-5.566, 8.152] / 1.493 [-5.157, 8.366]	0.017 [-0.797, 0.808] / 0.006 [-0.797, 0.824]	-0.018 [-0.353, 0.321] / -0.024 [-0.362, 0.317]
2	PPI	BN output gap	-0.228 [-0.697, 0.278] / -0.239 [-0.677, 0.240]	0.105 [-0.705, 0.909] / 0.100 [-0.713, 0.917]	-0.026 [-0.355, 0.313] / -0.029 [-0.345, 0.305]
3	PPI	Negative unemployment gap	0.078 [-0.329, 0.494] / 0.082 [-0.322, 0.489]	0.072 [-0.734, 0.902] / 0.065 [-0.761, 0.872]	-0.001 [-0.352, 0.338] / -0.004 [-0.338, 0.325]
4	CPI	Inverse markup	0.390 [-3.992, 4.804] / 0.652 [-3.743, 4.980]	0.041 [-0.636, 0.745] / 0.021 [-0.619, 0.690]	-0.088 [-0.350, 0.207] / -0.084 [-0.337, 0.192]
5	CPI	BN output gap	-0.091 [-0.370, 0.094] / -0.113 [-0.365, 0.071]	0.174 [-0.540, 0.912] / 0.151 [-0.534, 0.849]	-0.124 [-0.388, 0.187] / -0.106 [-0.359, 0.182]
6	CPI	Negative unemployment gap	0.111 [-0.106, 0.346] / 0.088 [-0.110, 0.311]	0.137 [-0.525, 0.817] / 0.105 [-0.551, 0.743]	-0.021 [-0.293, 0.247] / -0.022 [-0.284, 0.256]
7	Core CPI	Inverse markup	-0.206 [-4.042, 3.252] / 0.426 [-3.078, 3.597]	0.184 [-0.374, 0.705] / 0.131 [-0.357, 0.596]	-0.086 [-0.296, 0.158] / -0.060 [-0.254, 0.160]
8	Core CPI	BN output gap	0.051 [-0.106, 0.189] / 0.018 [-0.094, 0.119]	0.185 [-0.396, 0.721] / 0.101 [-0.356, 0.559]	-0.107 [-0.315, 0.152] / -0.054 [-0.240, 0.158]
9	Core CPI	Negative unemployment gap	0.091 [-0.092, 0.247] / 0.073 [-0.074, 0.204]	0.191 [-0.321, 0.666] / 0.121 [-0.341, 0.572]	-0.013 [-0.209, 0.215] / 0.004 [-0.182, 0.211]

Notes: QoQ uses the YoY endpoint dates. YoY aggregates the full quarterly design and uses $G_4 = A_4 A'_4$. A posterior for the QoQ-minus-YoY difference is deliberately not reported.

9 Persistent-error, low-frequency, and slow-only follow-up

The persistent-error specification uses a stationary AR(1) quarterly inflation disturbance. The low-frequency specification adds the pre-declared latent AR(1) inflation nuisance symmetrically across E0–E2. When the measurement gate fails, the slow-only specification is the reporting fallback rather than an ex post replacement for the visible E2 result.

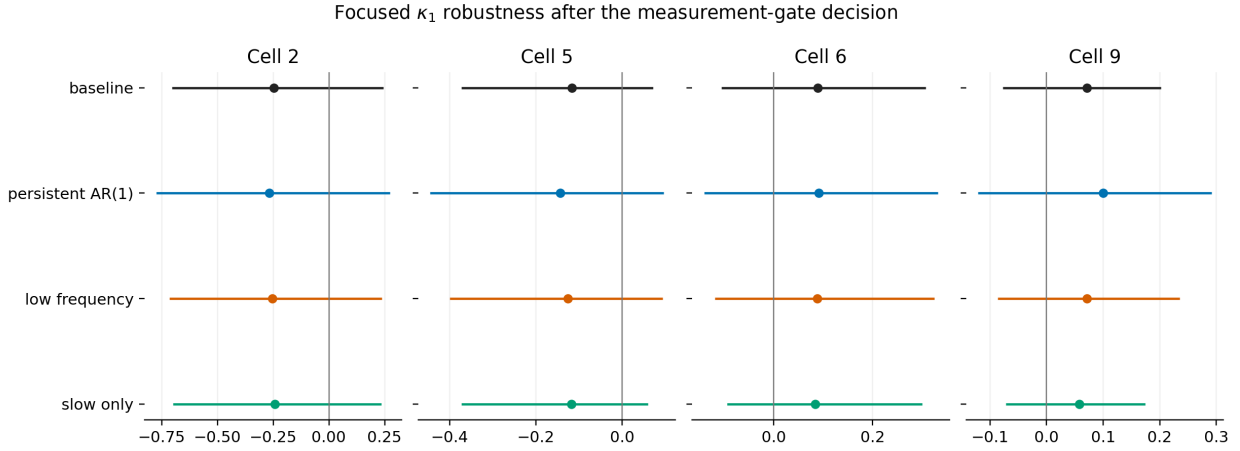


Figure 6: Focused κ_1 posterior summaries for Cells 2, 5, 6, and 9.

Cell	Specification	Mean	95% lower	95% upper	Sign probability
2	baseline	-0.246	-0.702	0.245	0.888
2	persistent ar1	-0.268	-0.774	0.274	0.870
2	low frequency	-0.253	-0.714	0.238	0.888
2	slow only	-0.242	-0.699	0.235	0.883
5	baseline	-0.117	-0.373	0.072	0.937
5	persistent ar1	-0.144	-0.445	0.097	0.937
5	low frequency	-0.126	-0.400	0.093	0.937
5	slow only	-0.117	-0.373	0.060	0.944
6	baseline	0.088	-0.106	0.308	0.867
6	persistent ar1	0.091	-0.141	0.332	0.839
6	low frequency	0.088	-0.120	0.325	0.837
6	slow only	0.084	-0.094	0.300	0.890
9	baseline	0.071	-0.078	0.202	0.938
9	persistent ar1	0.100	-0.122	0.292	0.905
9	low frequency	0.072	-0.087	0.235	0.891
9	slow only	0.058	-0.072	0.174	0.938

9.1 Global robustness-shift rule

Cell	Robustness	κ_1 shift (baseline SD)	Material	Decision
1	persistent ar1	0.12	No	Pass
1	low frequency	0.01	No	Pass
1	slow only	0.01	No	Pass
2	persistent ar1	0.09	No	Pass
2	low frequency	0.03	No	Pass
2	slow only	0.02	No	Pass
3	persistent ar1	0.06	No	Pass
3	low frequency	0.01	No	Pass
3	slow only	0.03	No	Pass
4	persistent ar1	0.05	No	Pass
4	low frequency	0.03	No	Pass
4	slow only	0.07	No	Pass
5	persistent ar1	0.23	No	Pass
5	low frequency	0.08	No	Pass
5	slow only	0.01	No	Pass
6	persistent ar1	0.03	No	Pass
6	low frequency	0.00	No	Pass
6	slow only	0.04	No	Pass
7	persistent ar1	0.41	No	Pass
7	low frequency	0.39	No	Pass
7	slow only	0.02	No	Pass
8	persistent ar1	0.22	No	Pass
8	low frequency	0.11	No	Pass
8	slow only	0.01	No	Pass
9	persistent ar1	0.39	No	Pass
9	low frequency	0.00	No	Pass
9	slow only	0.18	No	Pass

10 Economic magnitudes

Cell	Inflation	Activity	IQR-normalized slope contribution (pp)
1	PPI	Inverse markup	0.150 [-0.607, 0.940]
2	PPI	BN output gap	-1.146 [-3.271, 1.141]
3	PPI	Negative unemployment gap	0.397 [-1.845, 2.720]
4	CPI	Inverse markup	0.068 [-0.435, 0.552]
5	CPI	BN output gap	-0.543 [-1.737, 0.335]
6	CPI	Negative unemployment gap	0.490 [-0.588, 1.706]
7	Core CPI	Inverse markup	0.046 [-0.342, 0.406]
8	Core CPI	BN output gap	0.077 [-0.451, 0.532]
9	Core CPI	Negative unemployment gap	0.395 [-0.430, 1.121]

The table reports an IQR-by-IQR slope-change contribution in annualized inflation percentage points. It enables comparison in common outcome units without treating raw coefficients on differently scaled activity variables as comparable structural objects.

11 Auxiliary HSA benchmark

Item	Result
Status	conditional benchmark
Local QoQ endpoints	45
ζ_{ref} / b_x	6.00 / 1.00
δ_{HSA}	0.118 [-14.210, 14.567]
Equivalence band	± 1.599
Equivalence probability	0.171

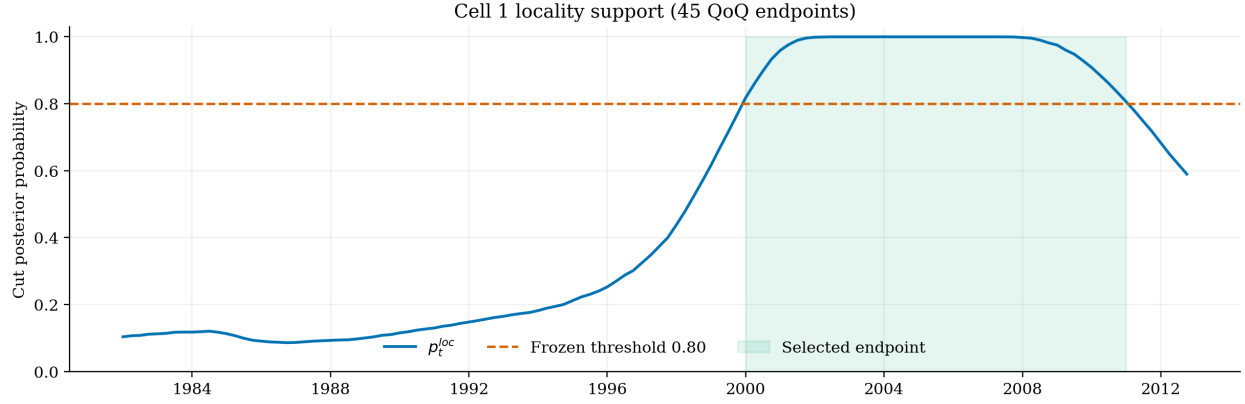


Figure 7: Frozen local-support rule based only on the measurement cut.

The benchmark is conditional on the unit-scale $b_x = 1$ calibration, maintained activity orthogonality, a GDP-deflator expectation proxy for PPI, and a calibrated reference markup. Compatibility is not a probability that HSA is true.

12 What this run does not establish

This run does not establish causal competition effects, a structural activity equation, real-time predictive validity, or a fully admissible PPI benchmark. Failed identification gates remain binding; unavailable external instruments, real-time vintages, and PPI-specific expectations are not treated as passed.